Percentiles and Z-Scores KEY

Name _____

You scored 85 on a test. Is that good? It depends on how everyone else did. Percentiles and z-scores tell where one value stands in a data set.

Percentile

The **percentile** of a value is the percent of the data that is **below** it.

$$\text{percentile} = \frac{\text{number of values below}}{\text{total number of values}} \times 100$$

Example: 20 students took a test. 15 students scored below 85.

$15 \div 20 = 0.75$, so a score of 85 is at the **75th percentile**. 85 is higher than 75% of the class.

Z-score

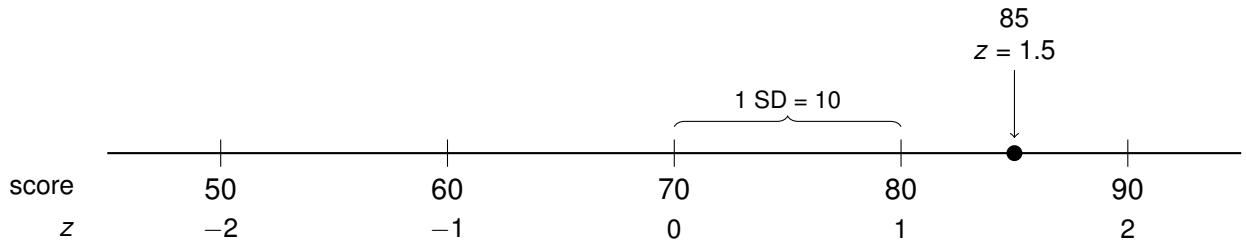
A **z-score** tells how many standard deviations a value is above or below the mean.

$$z = \frac{x - \mu}{\sigma} \qquad x = \text{the value}, \quad \mu = \text{the mean}, \quad \sigma = \text{the SD}$$

$z > 0$: above the mean. $z < 0$: below the mean. $z = 0$: equal to the mean.

Example: The test has a mean of 70 and an SD of 10. What is the z-score of 85?

$z = \frac{85 - 70}{10} = \frac{15}{10} = 1.5$ 85 is **1.5 standard deviations above** the mean.



μ is the Greek letter **mu**. σ is a small **sigma**. Do not mix it up with the capital Σ, which means add up.

Comparing with z-scores. Z-scores let you compare values from different data sets.

	Score	Mean	SD	z
Ana, math test	85	70	10	$\frac{85-70}{10} = 1.5$
Ben, history test	80	74	3	$\frac{80-74}{3} = 2$

Ana's score is higher, but Ben did better **compared to his class**. His z-score is larger.

Going backward: $x = \mu + z\sigma$ $z = -2$ on Ana's test: $70 + (-2)(10) = 50$

Work with a partner. Compare answers when you finish each problem.

1. Here are the quiz scores of 20 students, in order.

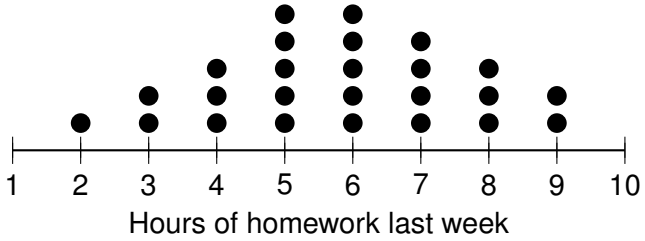
   52, 58, 61, 64, 66, 68, 70, 71, 73, 75, 76, 78, 80, 81, 83, 85, 88, 90, 94, 98

   (a) What percentile is a score of 83? 14 scores are below 83. $14 \div 20 = 0.70$, the 70th percentile.

   (b) What percentile is a score of 61? 2 scores are below. $2 \div 20 = 0.10$, the 10th percentile.

   (c) Which score is at the 50th percentile? 50% of 20 is 10 scores below it: 76.

2. This dot plot shows how many hours 25 students spent on homework last week.



   (a) What percentile is 7 hours? $1 + 2 + 3 + 5 + 5 = 16$ below. $16 \div 25 = 0.64$, the 64th percentile.

   (b) What percentile is 4 hours? 3 below. $3 \div 25 = 0.12$, the 12th percentile.

3. A class took a quiz. The mean is 20 points and the SD is 4 points. Find the z-score of each score. Then say what it means.

   (a) 26 $z = \dfrac{26 - 20}{4} = 1.5$. 1.5 SDs above the mean.

   (b) 14 $z = \dfrac{14 - 20}{4} = -1.5$. 1.5 SDs below the mean.

   (c) 20 $z = 0$. Equal to the mean.

   (d) 30 $z = \dfrac{30 - 20}{4} = 2.5$. 2.5 SDs above the mean.

4. Lin took two tests.

Test	Lin's score	Class mean	Class SD
Spanish	88	80	4
Science	90	84	6

   (a) Find Lin's z-score on each test. Spanish: $\frac{88-80}{4} = 2$. Science: $\frac{90-84}{6} = 1$.

   (b) Lin's science score is higher. On which test did Lin do better compared to the class? Explain.
   Spanish. Lin was 2 SDs above the mean in Spanish, but only 1 SD above in science.

5. A test has a mean of 70 and an SD of 8.

   (a) What score has a z-score of 2? $70 + 2 \times 8 = 86$

   (b) What score has a z-score of -0.5? $70 + (-0.5)(8) = 70 - 4 = 66$

6. In a 100-meter race, a **lower** time is better. The team's mean time is 14.0 seconds, with an SD of 0.5 seconds.

   (a) Maria ran 13.0 seconds. Find her z-score. $z = \frac{13.0 - 14.0}{0.5} = -2$

   (b) Is a negative z-score good or bad for Maria? Explain.
   Good. She is 2 SDs below the mean time, so she was much faster than most of the team.

7. A student says, "A z-score of -1 means the value is 1 less than the mean." Is the student right? Explain.

   No. It means the value is 1 **standard deviation** below the mean. With an SD of 10, that is 10 less than the mean.